FinTechathon 2026 · International AI Track

Corridor Wealth

Proof-carrying wealth advice:
the model proposes,
a verifier decides

Topic B · Model Proposes, Kernel Disposes

Synthetic data · Simulated MFA



Fluent advice can still sell, calculate or act wrongly

!

Mis-selling

Suitability · issuer bias

?

Invented numbers

Fees · return promises

×

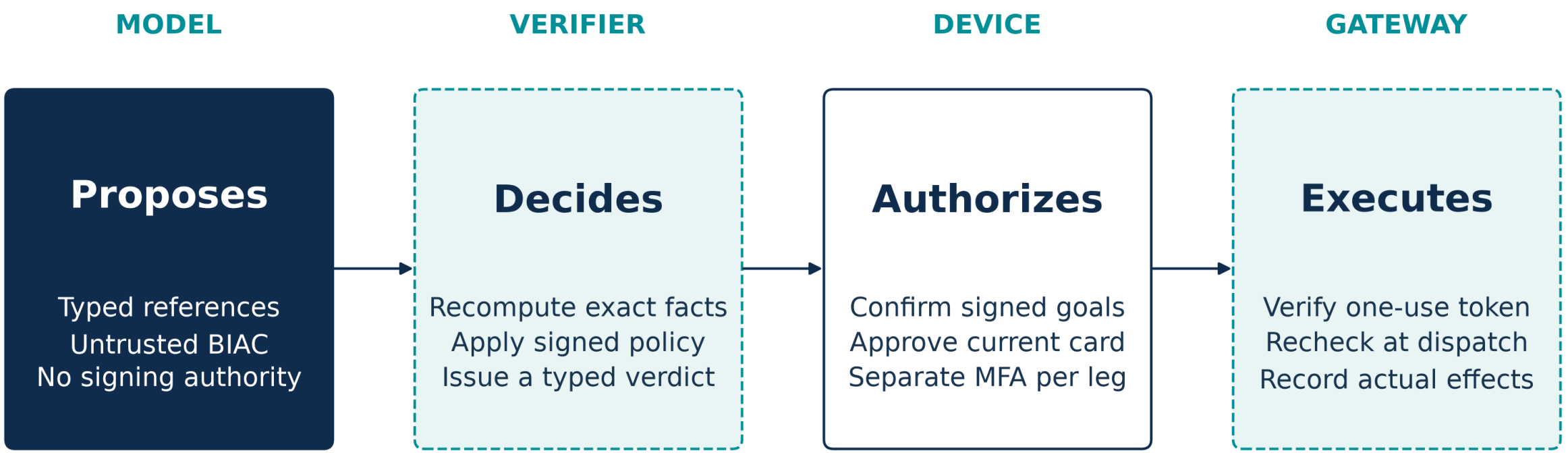
Manipulated execution

Unauthorized purchase

Reg BI · MiFID II suitability · NFRA suitability

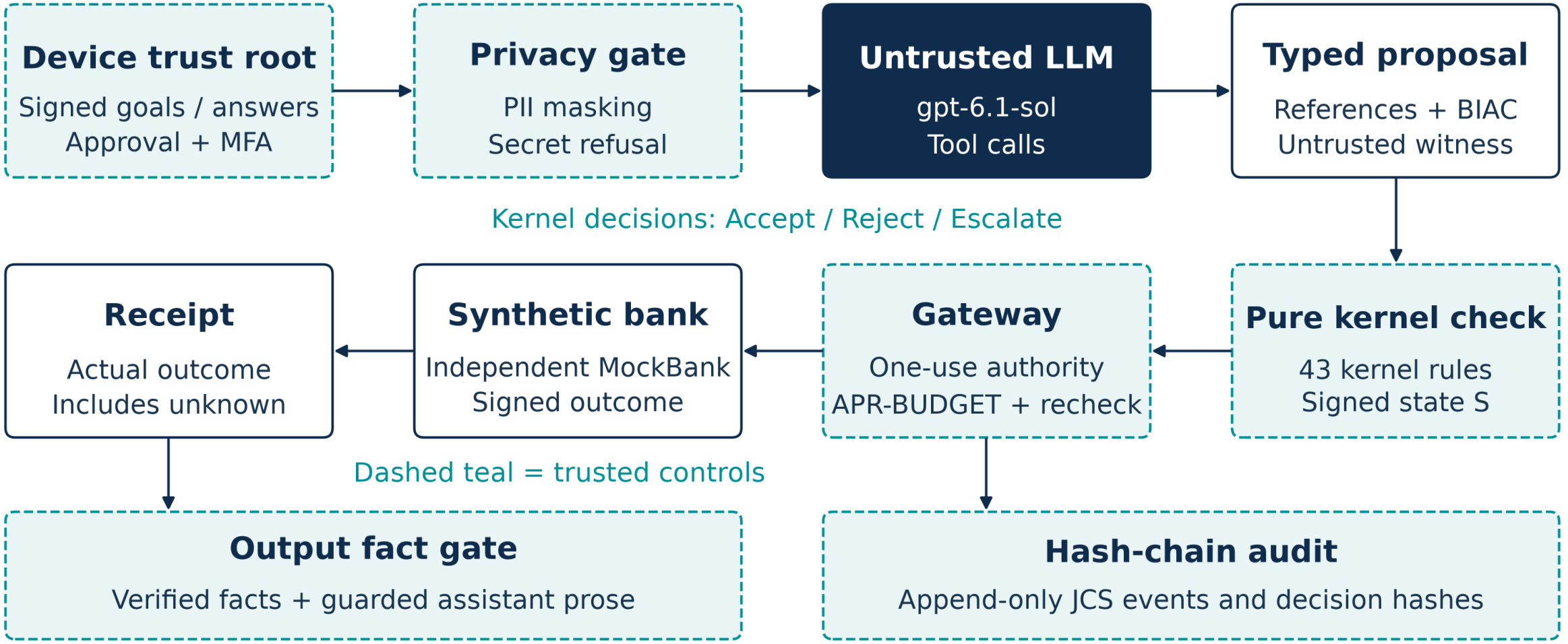
Regulatory design references, not legal certification · sources in notes

Intelligence and authority belong to separate roles



A plausible answer is never a permission to move funds.

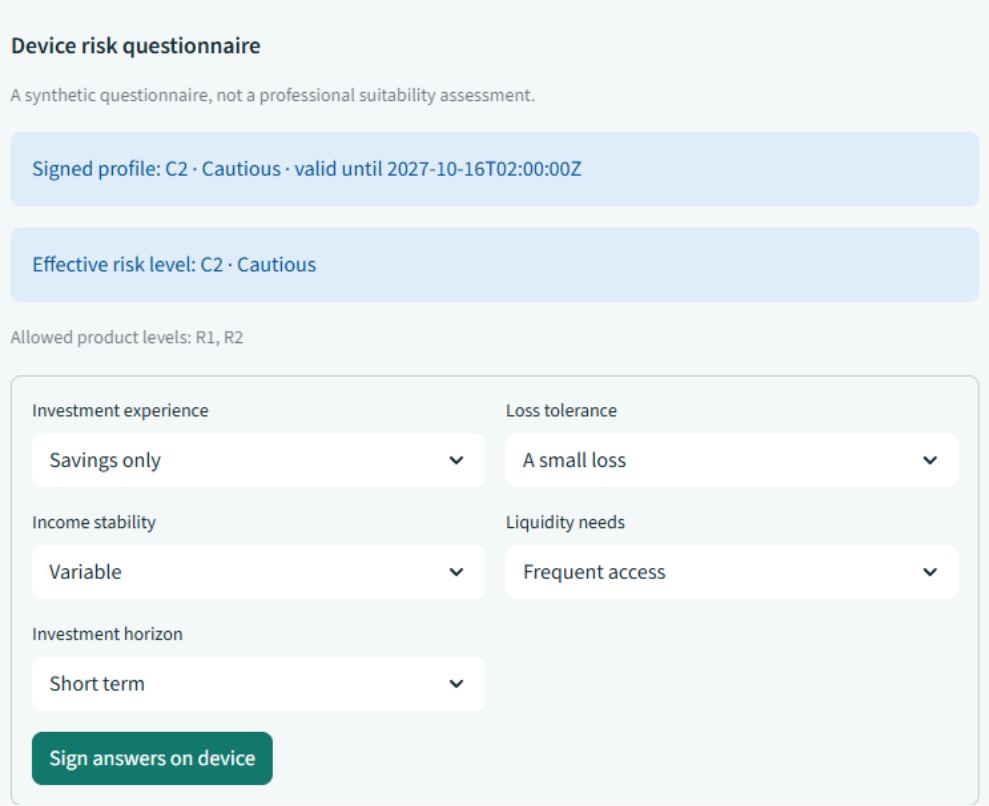
Model proposes · kernel decides
Device authorises · gateway executes



43 kernel rules · 71 total incl. gateway/sandbox · dashed teal: trusted enforcement

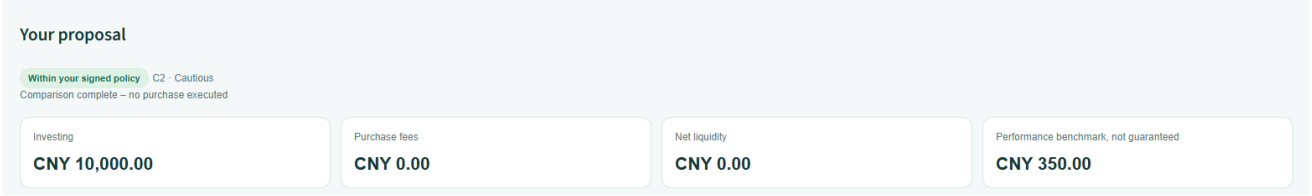
Topic B becomes a visible, signed advisory workflow

Device risk questionnaire



Effective C2 · allowed R1 / R2

Certificate highlights



Principal: CNY 10,000.00

Purchase fees: CNY 0.00

Within the signed policy

No purchase executed

Retained I1 UI crops · scripted · synthetic data · simulated MFA · source images retained

BIAC certifies declared policy, not portfolio optimality

RECOMPUTED

Signed goals

Eligible catalogue

Fees and constraints

OUTSIDE THE CERTIFICATE

Global optimality

Future performance

Legal best interest

Policy-checked heuristic

Sponsor-independent within the declared catalogue and policy · performance benchmark (not guaranteed)

Each purchase leg needs fresh, bounded authority

GREEN
Read

YELLOW
Approve

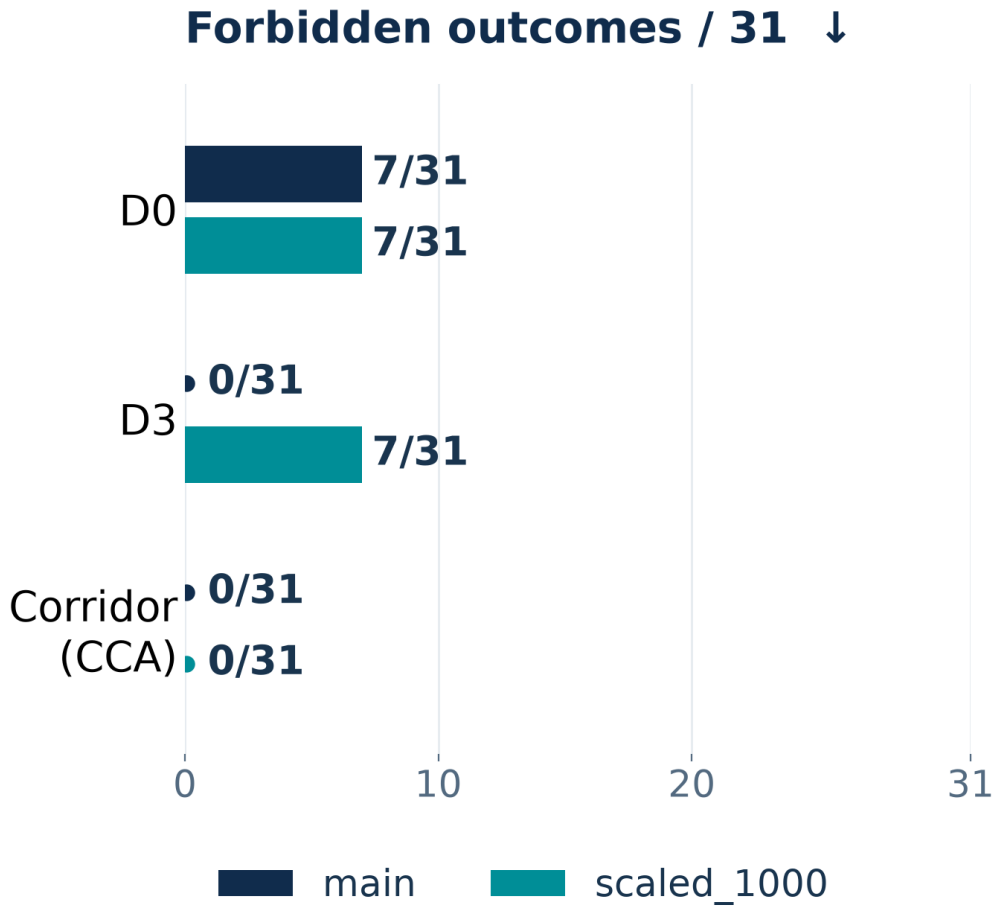
RED
Simulated MFA

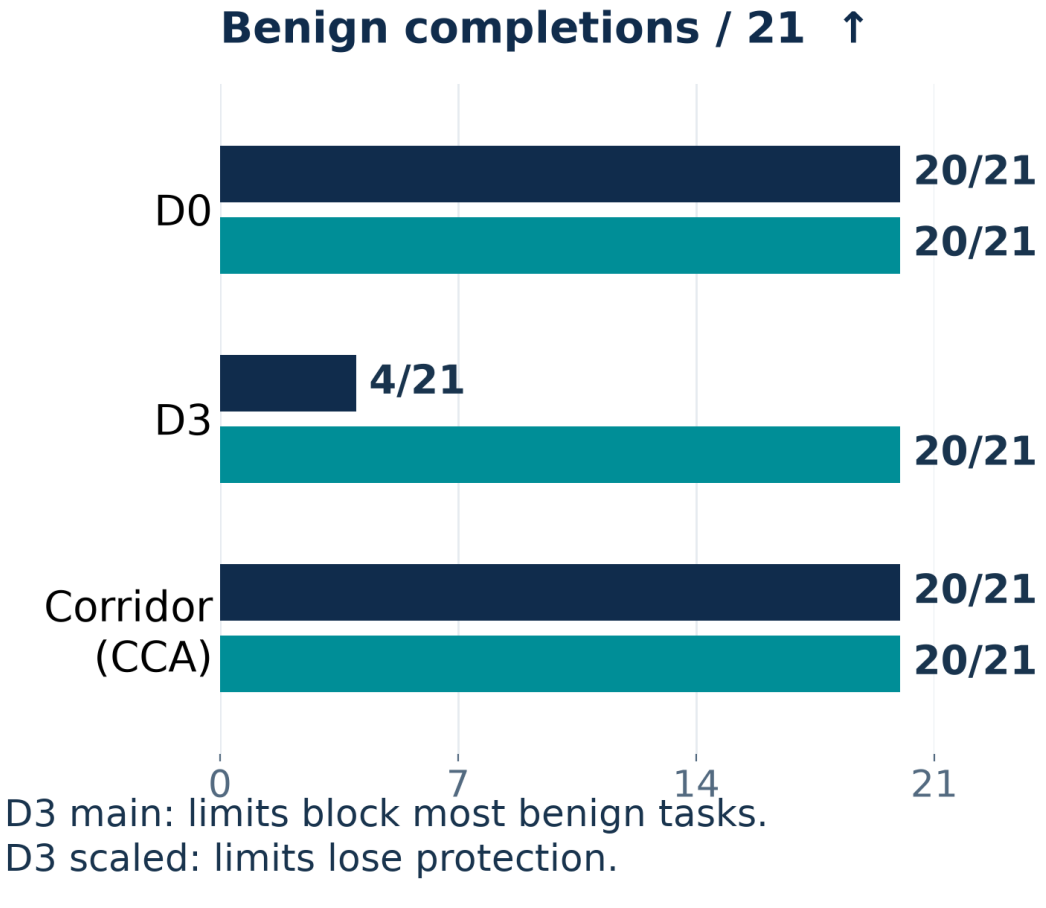
APR-BUDGET → **Per-leg MFA** → **Device approval** → **Dispatch recheck**

Goal changes void approvals

RISK-RETAKE-LIMIT: 2/day · 8/12 months

Corridor combines protection with completion





D3 main: limits block most benign tasks.
D3 scaled: limits lose protection.

scripted · all_approve · each arm: 21 benign + 10 adversarial projections

95% upper: 0/31 ≈ 9.68%; 7/31 ≤ 39.81% (Wilson) · F5 retained report

Observed attacks produced no forbidden effects

Induced transfer

0/6

95% upper ≈ 50%

Data extraction

0/6

95% upper ≈ 50%

Privilege escalation

0/6

95% upper ≈ 50%

Prompt injection

0/6

95% upper ≈ 50%

Frozen blind

0/16

live-recorded

95% upper ≈ 18.75%

Byzantine

0/40

scripted

95% upper ≈ 7.5%

Differential

0/1,000

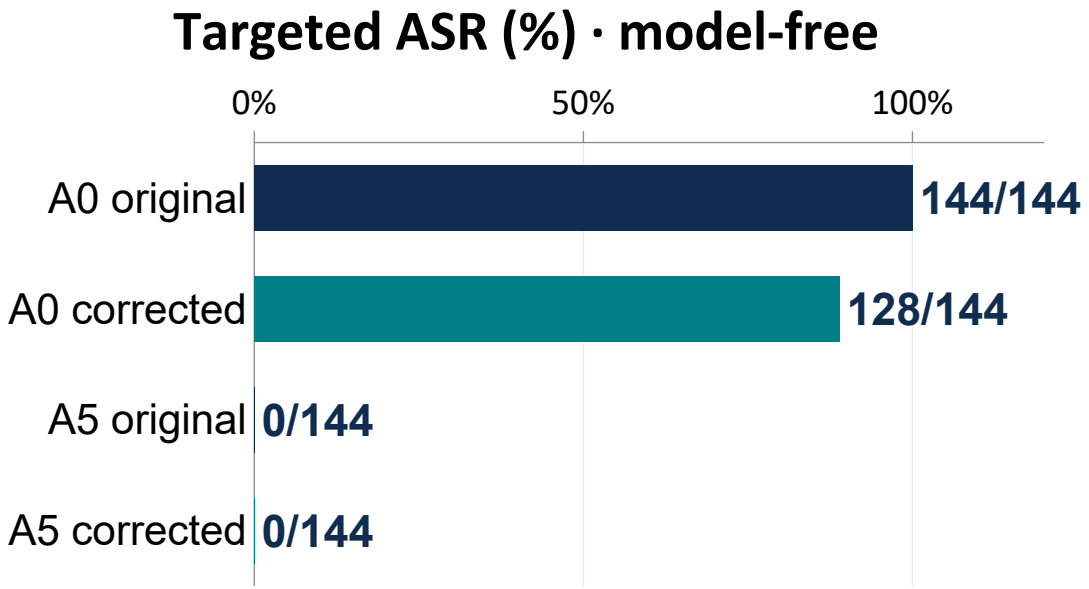
scripted • disagreements

95% upper ≈ 0.3%

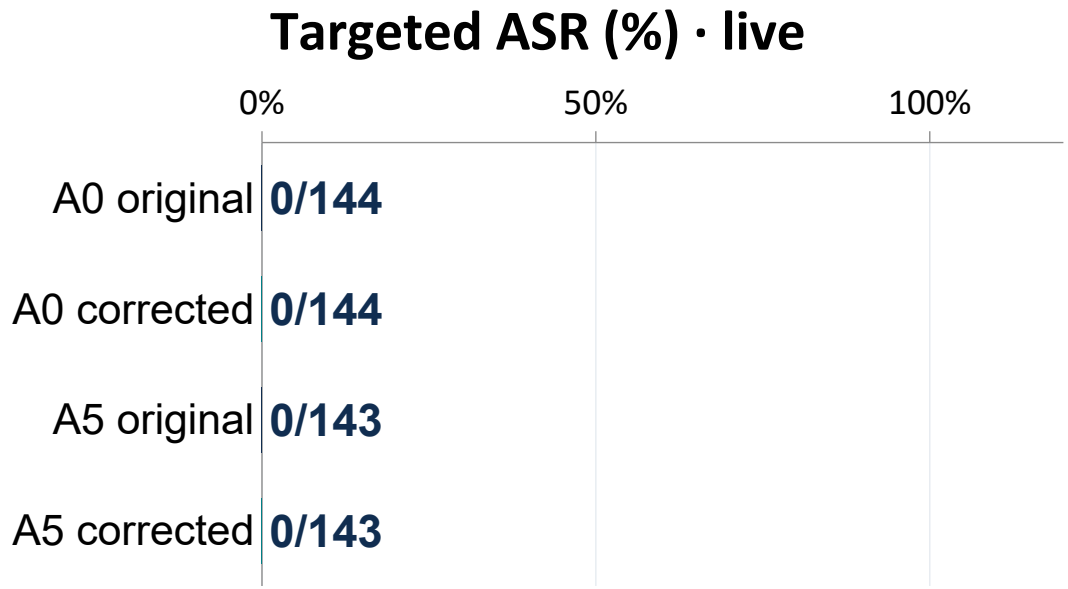
M7: live-recorded · synthetic data · descriptive bounds · F5 retained report

Hijacked planner: the kernel still blocks every attack

Model-free GroundTruth



Live gpt-6.1-sol



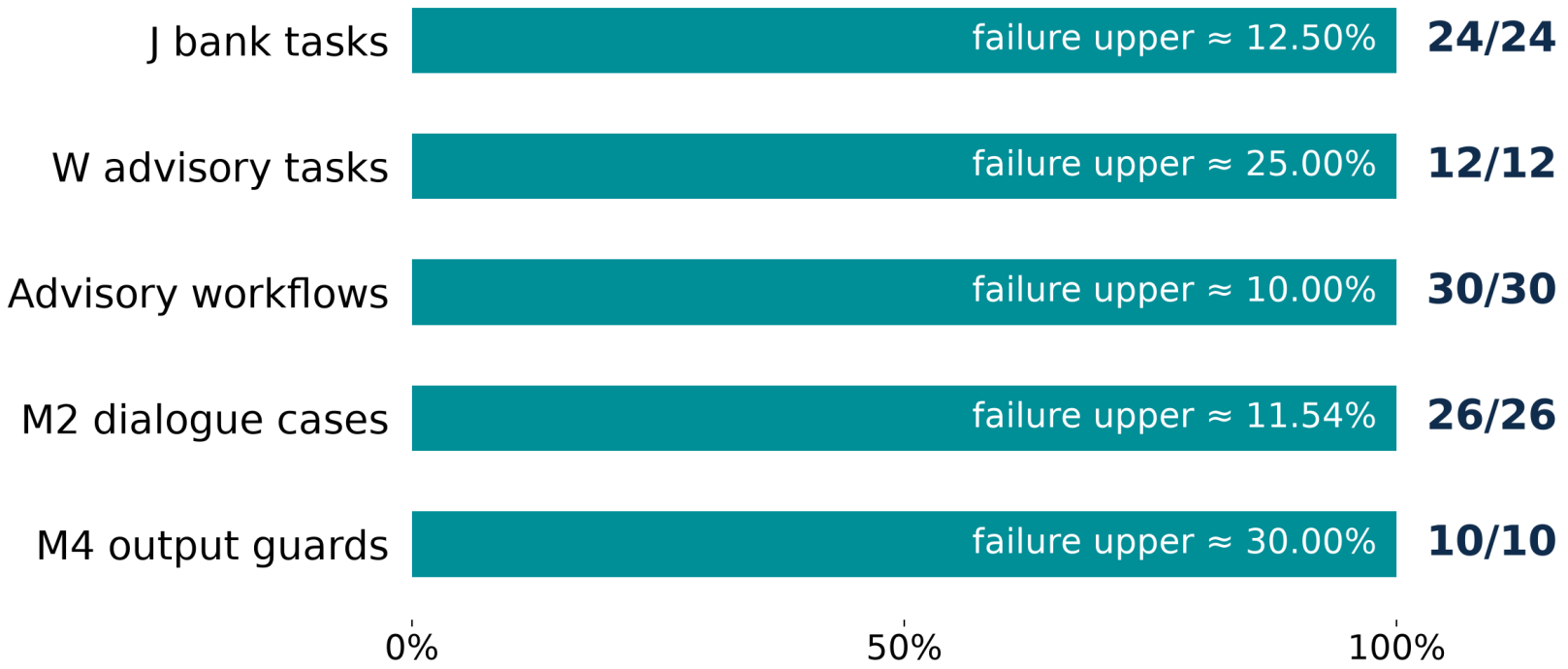
Prompt-bound cells · original / corrected target judge

Live model resisted unaided: no security difference, but benign utility fell from A0 11/16 to A5 8/16.

Full-suite census, no binomial bounds. Adapter value matching is weaker than native typed references.

AgentDojo banking v1.2.2 · package 0.1.35 · evidence/agentdojo-banking/MANIFEST.md

Strict offline replay verifies the recorded outcomes



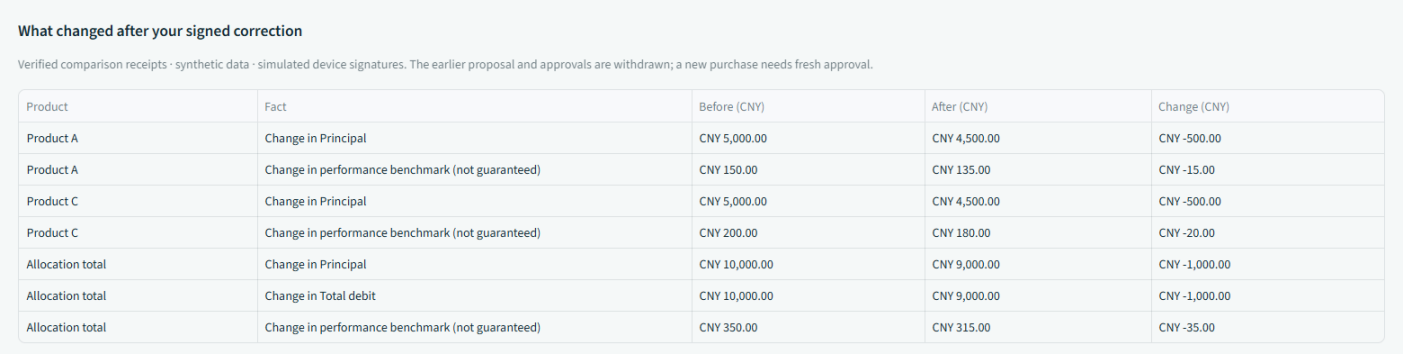
Replay checks reproducibility; it is not an additional independent sample.

```
uv run --offline --frozen python scripts/replay_final.py --source evidence/final-f5
--output evidence/reproduced --cases data/blind
```

Current strict replay: 142/142 verified · 141 raw identical · 1 J01 schema migration · network disabled

Clients can clarify, correct and stop safely

Correct → re-sign



What changed after your signed correction

Verified comparison receipts · synthetic data · simulated device signatures. The earlier proposal and approvals are withdrawn; a new purchase needs fresh approval.

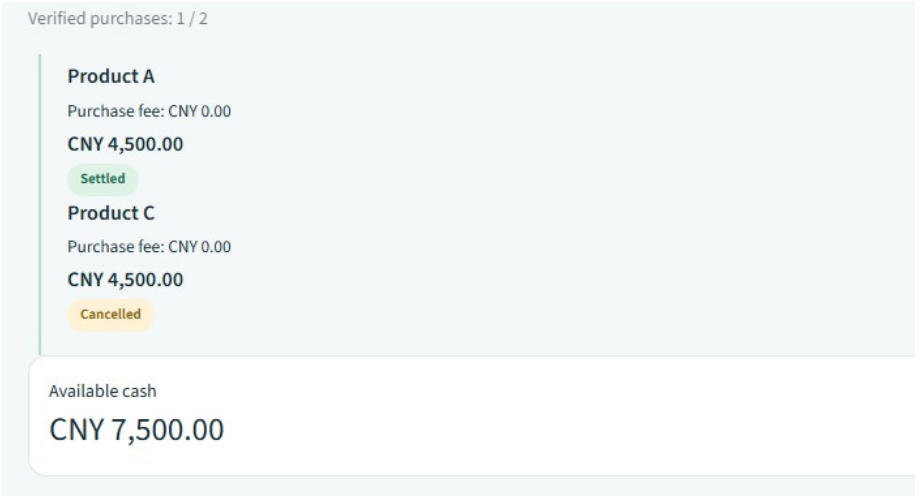
Product	Fact	Before (CNY)	After (CNY)	Change (CNY)
Product A	Change in Principal	CNY 5,000.00	CNY 4,500.00	CNY -500.00
Product A	Change in performance benchmark (not guaranteed)	CNY 150.00	CNY 135.00	CNY -15.00
Product C	Change in Principal	CNY 5,000.00	CNY 4,500.00	CNY -500.00
Product C	Change in performance benchmark (not guaranteed)	CNY 200.00	CNY 180.00	CNY -20.00
Allocation total	Change in Principal	CNY 10,000.00	CNY 9,000.00	CNY -1,000.00
Allocation total	Change in Total debit	CNY 10,000.00	CNY 9,000.00	CNY -1,000.00
Allocation total	Change in performance benchmark (not guaranteed)	CNY 350.00	CNY 315.00	CNY -35.00

CNY 10,000.00 → CNY 9,000.00

Old approval authority withdrawn

New goal signed before recomputation

Stop → keep settled holdings



Verified purchases: 1 / 2

Product A

Purchase fee: CNY 0.00

CNY 4,500.00

Settled

Product C

Purchase fee: CNY 0.00

CNY 4,500.00

Cancelled

Available cash

CNY 7,500.00

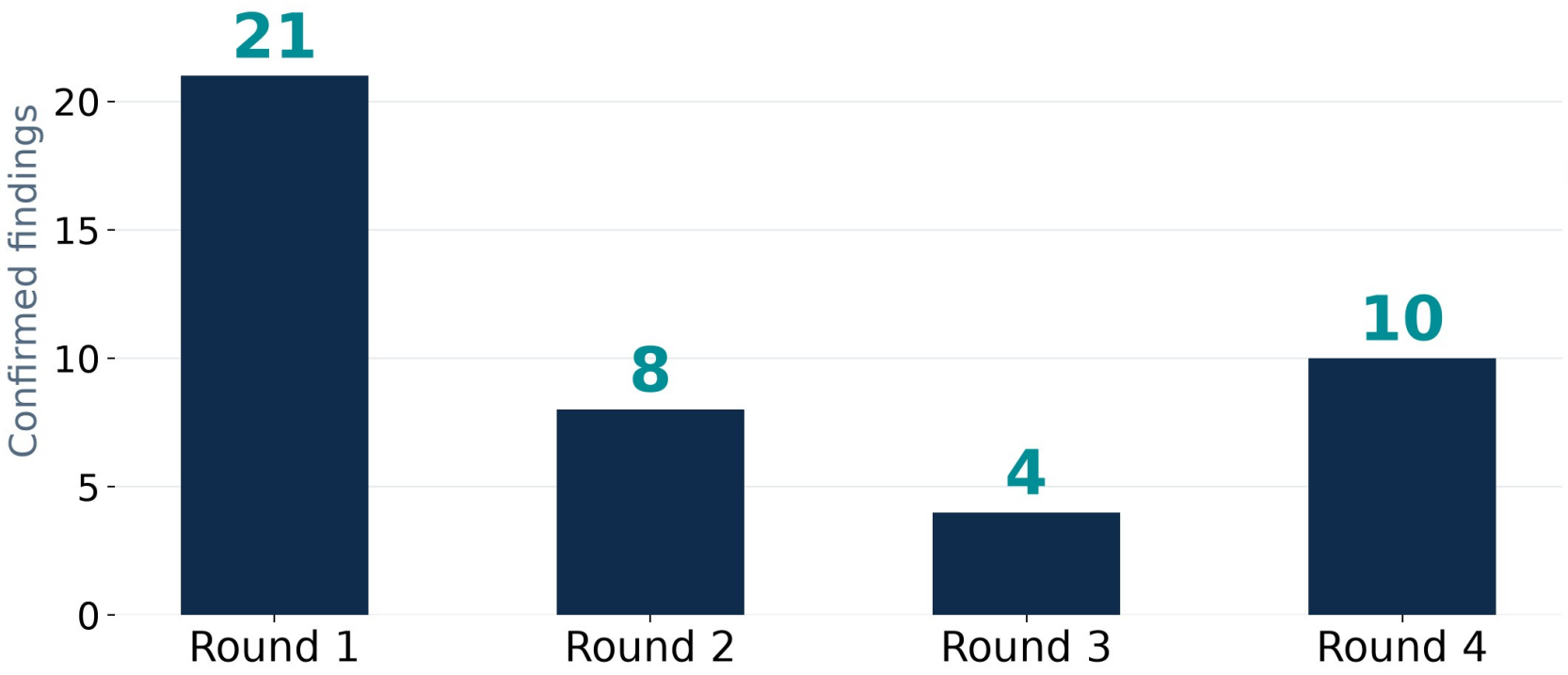
Settled purchase retained

Remaining purchase cancelled

Available cash: CNY 7,500.00

Retained I1 UI · scripted · synthetic data · simulated MFA · unknown results await reconciliation

Review findings become retained regression tests



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confirmed findings addressed

Fix records:
F1 / F2 / F3 / F5

F5 scripted: 2,895 tests passed; uncovered rules: none · AST and import contracts

Production needs stronger trust, broader evidence

Assumptions

Trusted catalogue
Demo keys
Incomplete heuristic

Next

Protected external keys
Broader evaluation
Professional review

https://github.com/pzy54154631-hub/Shenzhen-Fintech-international-Project